

Homework 2: Middle-Ear Simulation Extension

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Website Link: <https://interactive-ear-modeling.vercel.app/>

Github Link: <https://github.com/mml260/Interactive-Ear-Modeling>

Objective and Model Scope

This portion of the build extends upon the outer ear simulator with a standalone web application model. The signal path of the model goes from the pinna and concha, to the ear canal, the eardrum, ossicles, and ends with the cochlear load. The app displays and provides real time audio playback for the outer ear, middle ear, and both of them combined. Bascading between the outer ear model and inner ear model is connected as a multiplication of their transfer functions ($H_{total}(f) = H_{outer}(f) \times H_{middle}(f)$) but doesn't account for any loading between the stages.

Figure 1: Outer, middle, and combined magnitude responses in the web application.



Middle Ear Circuit Verification

The circuit representation of the middle ear that was provided was implemented as subsystems including Z_{cavity} , $Z_{eardrum}$, $Z_{ossicles}$, Z_{joint} , and $Z_{cochlea}$. The output transfer function is the pressure ratio from the input to the cochlear load. The impedances and subsequent transfer function are listed below.

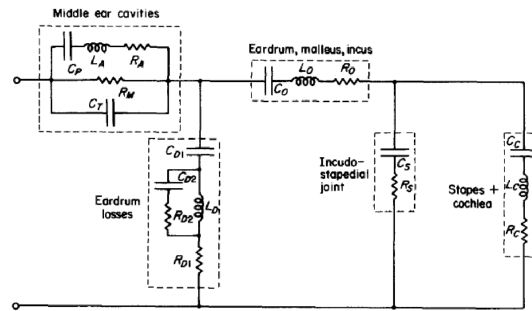
$$\begin{aligned} Z_{cavity} &= (Z_{Cp} + Z_{La} + Z_{Ra}) \parallel Z_{Rm} \parallel Z_{Ct} \\ Z_{eardrum} &= Z_{CD1} + (Z_{LD} \parallel (Z_{CD2} + Z_{RD2})) + Z_{RD1} \\ Z_{ossicles} &= Z_{CO} + Z_{LO} + Z_{RO}; \quad Z_{load} = Z_{joint} \parallel Z_{cochlea} \\ Z_{middle} &= Z_{cavity} + [Z_{eardrum} \parallel (Z_{ossicles} + Z_{load})] \\ H_{middle} &= (Z_{after_A} / Z_{middle}) \times (Z_{load} / (Z_{ossicles} + Z_{load})) \end{aligned}$$

The function's validity was proved in code rather than comparing output graphs. At 1 kHz, the tests proved each branch from the impedance equations and compared the real and imaginary components of each subsystem. The tests additionally verified that the resistors impedance does not change as the frequency changes and that voltage and current stay in sync. As frequency goes up, the inductor's impedance goes up therefore blocking more current. The capacitor impedance goes down as frequency increases, allowing more current to flow and the current is 90 degrees out of phase with the voltage. 21 automated tests were run prior to the production build.

Component Values

The exact component values for the given circuit in Figure 2 were not provided, so the values were derived from a Zwislocki style baseline.

Figure 2: Zwislocki style circuit diagram of middle ear



From Zwislocki's 1962 publication "Analysis of the Middle Ear Function. Part 1: Input Impedance" (<https://www.yorku.ca/cberge/1410W2013/lnotes/Zwislocki1962.pdf>), the cavity impedance and compliance was derived from the frequency response provided from figure 3 on page 1516 and the simplified malleus and ossicle circuit behavior is explained on page 1517. Baseline component values were transcribed from Zwislocki's historical electrical analogue of middle-ear input impedance. These aggregate values were derived and fitted using normal and pathological ear impedance measurements together with anatomical cavity estimates. They are used as a reproducible baseline rather than patient-specific measurements. The cochlear inertance was set to zero as a documented model simplification.

Figure 3: Replaceable component values in the model

Advanced model options		Source-tagged values
Replaceable Zwislocki-style baseline. Values are retained in SI units inside the model.		
L_A	14 mH · cavities	
R_A	1000 Ω · cavities	
C_p	5.1 μ F · cavities	
R_m	60 Ω · cavities	
C_t	0.35 μ F · cavities	
C_{D1}	0.23 μ F · eardrum	
R_{D1}	40 Ω · eardrum	
C_{D2}	0.4 μ F · eardrum	
R_{D2}	220 Ω · eardrum	
L_D	15 mH · eardrum	
C_O	1.4 μ F · ossicles	
L_O	40 mH · ossicles	
R_O	70 Ω · ossicles	
C_s	0.25 μ F · joint	
R_s	3000 Ω · joint	
C_C	0.6 μ F · cochlea	
L_C	0 mH · cochlea	
R_C	600 Ω · cochlea	

Three AI-Assisted Experiments

Each case listed on the middle ear section of the model changes one perimeter of the middle ear while resetting the other controls to the OME scenario. The predictions were made before running the model and are compared with the calculated ear response.

Experiment 1 - Ossicular Inertance

Controlled change: LO: 40 → 60 mH (+50%).

Pre run prediction: A 50% increase in inertance should attenuate high frequency after middle frequency peak.

Calculated result: Peak frequency moved 871 Hz to 752 Hz. Relative to baseline, the response changed -1.68 dB at 2 kHz and -2.95 dB at 4 kHz.

Comparison: The high movement of the low pass filter occurred as expected. However, it was noticed that there is a slight peak increase for the broad band frequency peak.

Figure 4: Inertance comparison



Experiment 2 - Eardrum Compliance

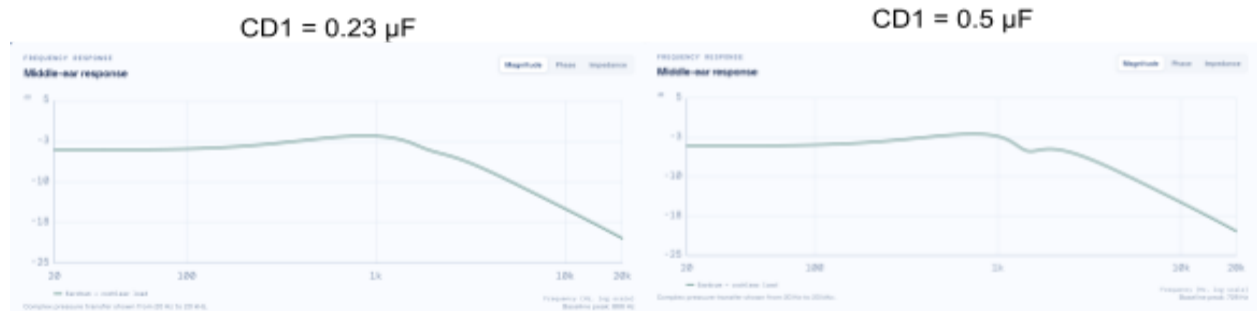
Controlled change: CD1: 0.23 → 0.50 μ F

Pre run prediction: A higher capacitance should change the behavior of the broad peak by attenuating it.

Calculated result: Peak frequency moved from 871 Hz to 32 Hz and peak magnitude changed from -1.49 to -1.82 dB. The response at 1 kHz was -0.73 dB relative to baseline.

Comparison: The response moved downwards and was consistent with the prediction.

Figure 5: Compliance Comparison



Experiment 3 - Ossicular Loss

Controlled change: R0: 70 Ω $\rightarrow$ 300 Ω

Pre run prediction: Higher resistance should cause more energy to dissipate which would reduce the amplitude of the broad peak

Calculated result: Peak frequency moved from 871 Hz to 414 Hz. The peak magnitude changed from -1.49 to -3.83 dB. The response at 1 kHz was -2.69 dB relative to baseline.

Comparison: As predicted, the larger the resistance, the peak lost significant magnitude.

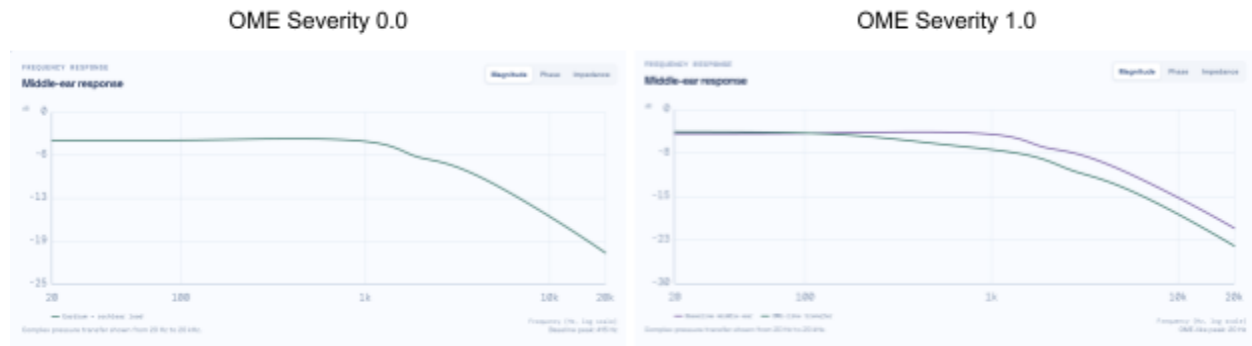
Figure 6: Loss Comparison



OME Transfer Scenario

The app includes a small child otitis media with effusion (OME) sensitivity scenario. Its continuous severity control reduces eardrum and joint compliance while increasing ossicular inertance and loss. When severity is above zero, the app overlays the baseline and OME responses and applies the same profile to playback.

Figure 7: Small child OME transfer comparison



AI Use and Engineering Log

For the circuit interpretation and component values, I decided to use the verified topology that was provided in the notes instead of using AI to interpret alternative routes that aligned with the frequency response from the Zwischel paper. As a result, this also confirmed that the values that were chosen retained a similar baseline rather than coming up with arbitrary values. When building the app, the first custom FFT filtering approach failed causing real time playback errors and glitching in the audio playback. This resulted in replacing it using the [fft.js](#) library which resulted in accurate frequency response and audio playback.

Limitations

The model is a first order approximation of the human ear, it is not a completely accurate anatomical model and cannot be treated as such. Additionally, the values that were chosen were derived using historically used numbers and to get close to the Zwischel style baseline but are limited in the fidelity and not measured for a specific listener. Additionally, there is no loading between stages and the transfer functions are multiplied which doesn't represent the actual anatomy and impedance change of the ear. Lastly, the OME scenario is not clinically calibrated and is not to be used as a diagnostic model.